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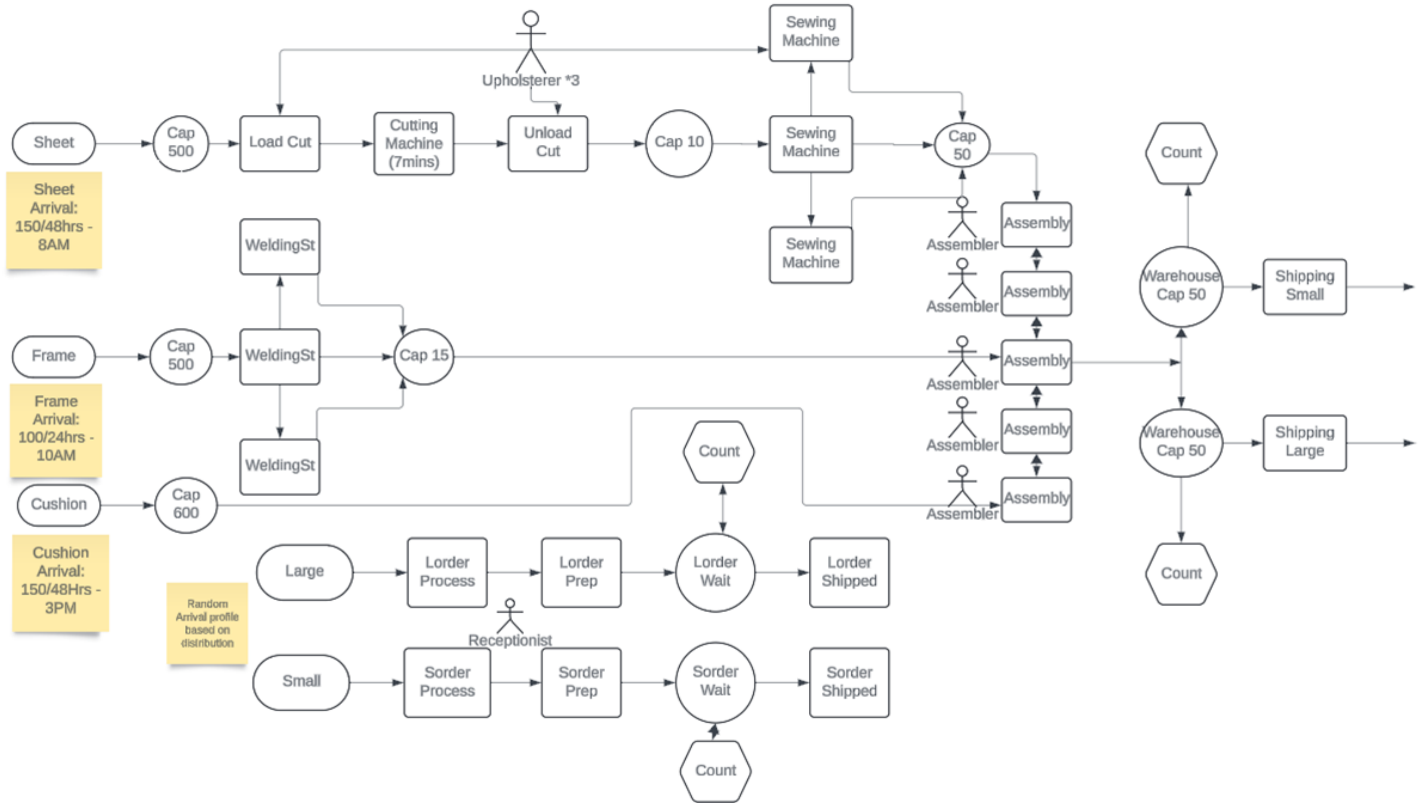
Question 1

To develop an advanced functional model, the model must be conceptualized through a flow chart to give an idea of how the model would be presented on witness. Following this, a flow chart has been drawn up to show exactly how the model itself would look and work.

Firstly, the scope of the model must be identified. These are the key areas essential for a comprehensive understanding of the plant's operation.

- Inventory management: Simulating fabric sheet, frame kit and cushion deliveries and their storage capacities to ensure that the model can accurately reflect the logistical challenges and constraints of the simulation.
- Manufacturing processes: The Cutting, welding, sewing and assembly processes; each with defined durations and capacities to mirror actual operations as closely as possible
- Assembly workforce: The conceptual model considers the human resources available, which are three upholsterers working with the fabric sheets, two cutting machines, three sewing machines, three welding stations, five assemblers and five assembly stations, a receptionist to process orders, and two workers to manage the queues for small and large orders. This ensures all workers are accounted for and considered, and the model itself is accurate.
- Order fulfillment: The model simulates the storage of chairs in warehouse sections for small and large orders, handling and processing of incoming orders, collection of chairs for shipping. This part focuses on the final steps of the model and the order-to-delivery process, making sure that customers' orders are fulfilled efficiently and accurately.



The conceptual model shown above gives a detailed overview of the chair manufacturing process, starting with the introduction of three key components: fabric sheets, metal frames, and cushions, each with specific arrival patterns based on predefined data. Firstly, fabric sheets first enter a buffer with a capacity of 10, then proceed through three machines dedicated to loading, cutting, and unloading, each operating with specific cycle times. After these stages, the cut sheets are moved to another buffer designed to hold up to 50 sheets until an upholsterer is available for sewing. Simultaneously, metal frames are taken from their initial buffer to one of three welding stations, each guided by an exact cycle time distribution, and upon completion, the frames are stored in a buffer with a capacity for 15 finished frames until needed for assembly. Cushions, already prepared for assembly, bypass any processing and go directly to the assembly buffer. The assembly phase involves five machines, each operated by one worker, where assembly commences only when each machine receives one unit of fabric, frame, and cushion from their respective buffers. After assembly, the chairs are directed to one of two warehouses, chosen based on which has more capacity available at the time. In parallel to the manufacturing process, the model simulates order intake and processing, categorizing orders as large or small, and managing them through respective processing units, Lorder process for large orders and Sorder process for small orders, both overseen by a single receptionist. Once orders are processed, they move to a preparation stage, Lorder preperation for large and Sorder preparation for small orders, which simulates the preparation time, and then to waiting buffers equipped with a counting mechanism to alert the warehouse when an order is ready to be shipped. If the warehouse has chairs available, the order is completed and the chairs are dispatched, ensuring a streamlined process from raw material intake to finished product shipping.

The following are various assumptions and simplifications made in relation to the conceptual model,

- Continuous operation: Staff work with no interruptions or delays within the working hours, excluding lunch breaks
- Workforce reliability: Machines do not break down; staff are always available and never absent.
- Consistent process times: Manufacturing processes always take the same amount of time/always follow the given distributions.
- Reliable delivery: Deliveries of materials are always consistent and on time.
- Customer orders cannot be cancelled or modified after they have been made
- No faulty chairs are produced; therefore, no returns are made (this will be modified later)
- Market demand constantly follows a given distribution, without considering trends or seasonality.
- Infinite order queue capacity, with no orders lost due to queue overflow
- All processes are simply paused and resumed at breaks or at the end of a shift, with no handover time or loss of efficiency.

The following points represent the outputs from the simulation will be studied as metrics for the model to measure its performance:

- Average times between orders being placed and getting shipped. This would measure the responsiveness of the production system and its ability to meet customer demands.
- Percentage of orders being shipped within 24 hours. This is a critical measure of the level of service and speed of delivery, giving an estimate of how well the plant manages orders.
- Utilization rates of staff and machines. This helps understand the efficiency of resources and identify any bottlenecks.

- Levels of inventory for fabric sheets, frame kits and cushions. Monitoring stock levels to ensure there is sufficient material to meet production needs without overstock

These metrics will be used to see whether the objectives are being achieved, and to identify the reasons why they are/aren't. By adjusting the model, changes in these outputs will be observed, to find optimal solutions for bottlenecks/other issues in the manufacturing plant.

Since the metrics have now been covered, we can now introduce the distributions used for each element of the model. The following distributions were used for processes which could take anywhere between two given amounts of time. Other processes, such as the cutting machine, always took exactly 7 minutes, so a distribution was not needed for these. The distributions have been selected with help from Stat::Fit- collections of data from the past on the time taken for each process and was taken to find the best suited distributions for each process's time. From these collections of data, Stat::Fit was used to find the most appropriate distributions which fit the historical data the best. These were then used as the cycle times in the Witness model, to most accurately simulate the time taken for certain components of the manufacturing plant. All distributions are approximate to two decimal places.

To help give an idea of how the distributions were found, we will show you the Stat::Fit results for each distribution. After inputting the outside information provided to us into Stat::Fit, we see the following:

autofit of distributions

distribution	rank	acceptance	aicc prob
Gamma(0, 2.74, 11.2)	100	do not reject	0.831
Beta(0, 2.04e+003, 2.7, 178)	96.6	do not reject	0.295
Erlang(0, 3, 10.2)	94.7	do not reject	1
Lognormal(0, 3.23, 0.671)	17.1	do not reject	0
Rayleigh(0, 25.3)	0.384	do not reject	0.0257
Pearson 5(0, 1.89, 35.9)	0.000416	reject	0
Exponential(0, 30.6)	0	reject	0
Triangular(0, 97.4, 12.7)	0	reject	0
Uniform(0, 96.4)	0	reject	0
Chi Squared(0, 26.2)	0	reject	0
Power Function(0, 97.1, 0.742)	0	reject	0
Weibull	no fit	reject	0
Pearson 6	no fit	reject	0

From this screenshot, we can see the autofit of distributions for Assembly time in our simulation model, and which of those are recommended and should be used/rejected. The

'Rank' column highlights, out of 100, how accurate the distribution is. The distributions are either accepted or rejected based on their fit to the dataset, and Stat::Fit suggests which are the most compatible distributions. In this case, a gamma distribution is the most fitting for the data we have been provided with, so for the distribution shown above we know the distribution we need to use in our model for Assembly Time is:

gamma (2.74, 11.20)

This process was repeated for every other distribution used throughout the model. Following is a list of each process used, and the appropriate distribution chosen by Stat::Fit as the best fir for each different dataset, which was provided to us by Congress:

- Loading cutting machine- LogNormal(1.45,0.52)
- Unloading cutting machine- erlang (1.49,3.00)
- Sewing machine- LogNormal (9.06, 5.79)
- Assembly time- gamma (2.74, 11.20)
- Welding time- LogNomral (14.91, 1.01)
- Large order daily arrival profile- normal (2.37, 2.50)
- Small order daily arrival profile- LogNormal (24.10, 9.94)
- Order processing time- weibull (3.68, 6.55)
- Large order preparation time- LogNormal (9.46, 6.93)
- Small order preparation time- gamma (5.14, 0.20)

All the processes use the optimal distribution for their dataset, and we know this because Stat::Fit guarantees us high precision of each distribution for each process.

Question 2

The model created in Question1 is validated through various procedures which ensure the model's internal logic and external applicability are examined. This ensures that the simulation generates accurate and realistic forecasts to help guide effective decision making. The model drawn up in Question 1 must be critically assessed to ensure it accurately reflects the operations of the manufacturing plant. It must then be analyzed to ensure that every aspect, presumption, and model simplification is valid and warranted. Since these elements have an impact on the realism and usefulness of the simulation, their suitability is vital to the effectiveness of the model developed. The process of Validation is explored through White Box Validation, Black Box Validation, and Data Validation.

White Box Validation, which tests the internal structures and working of the model, validates the model designed as follows.

> Logical Testing – Comparing the values deduced by the model to information that has been provided, and showing their similarities, to ensure accuracy throughout the model. Each of the model's outputs is checked for consistency and accuracy before integrating it into the simulation. Every one of the manufacturing times and order preparation times and daily orders given in the information provided match effectively with the model. We compared the values our conceptual model generated to real values provided to us, and combined they helped provide an accurate simulation.
>
> Code Verification – The model was developed in many steps and ran repeatedly after each new element was added to ensure the model was always valid and was not flawed. Making sure the model runs properly with no errors and through controlled conditions to see if the outputs logically reflect the inputs provided. To ensure the model was as accurate as possible, we repeated the model itself and changed the process we were using at least 15 times, with changes to each document to make sure our simulation was as accurate as possible.

Black Box Validation are tests carried out to evaluate the system's external behavior. They can help validate the model as follows.

> System Behavior Testing – Similar to the Logical Testing, comparing the overall model output with the real-world data from the manufacturing plant to check the accuracy of the model developed. This involves statistical tests to see, if any, significant differences between the simulated results of the model and actual historical data.
>
> Scenario Analysis – The model operates on data distributions given from real world examples. Therefore, the model can analyze unpredictable and anomalous data. Due to this, the model can react when greater orders are required. Our distributions monitoring the arrival of large and small orders are well-supported by extensive data, allowing for adjustments to accommodate variations in order size.

Data Validation ensures the accuracy and reliability of the data used throughout the simulation, and validates the model as follows;

Sensitivity Analysis – this was conducted to understand the influence of individual factors and their interactions throughout the model. Buffer capacity, arrival time distributions, and arrival rates were varied independently and in combination. This analysis aimed to identify the model elements with the most significant impact on system performance.

Data Source Verification – all the data inputs in the model are verified for accuracy and relevance, to ensure they are playing a role in the model. This also includes checking data sources for consistency and reliability. To check our data sources' reliability, we use Stat::Fit to check our sources, and Excel to keep all our values calculated in one place, as will be seen in Question 5.

The procedures used throughout the model are comprehensive and designed to ensure that every aspect of the model is analyzed for accuracy and reliability. Through a combination of both White and Black Box validation, alongside Data Validation and running controlled experiments to further validate the model, the simulation is fine-tuned to provide insights that can guide operational improvements and strategic decision-making.

3)

After simulating Congress Office Chair Plants production operation, we have found some elements can be changed to the inventory deliveries to optimize operation chain and costs.

Currently, the delivery schedule for the chair parts is as follows:

- Fabric sheets- 8am, every two days, 150 sheets
- Frame kits- 10am, every day, 100 kits
- Cushions- 3pm, every two days, 150 cushions

This delivery schedule means that there are 56 separate deliveries being made every 28 days. Each shipment has a fixed charge regardless of how many items are delivered, so reducing the number of deliveries would reduce the overall cost, and therefore improve the model's efficiency. To do this, an alternative delivery schedule plan has been made:

- Fabric sheets- 8am, every 7 days, 500 sheets
- Frame kits- 8am, every 7 days, 500 kits

- Cushions- 8am, every 9 days, 600 cushions

This came from analyzing the output of the model, on both slow days and busy ones to ensure we will have enough stock in no matter what. (on average we can produce 60 chairs a day). This plan fills up the inventory of each part every week, except cushions due to them having a larger inventory. This schedule is more efficient, as more parts can be ordered less frequently. With this schedule, 11 deliveries will be made every 28 days. With this plan, delivery costs would be cut by over 80%, while still receiving adequate amounts of parts.

Data was collected for the mean time between an order being placed and the items being shipped (for both small and large orders); and the percentage of orders shipped within 24 hours of being placed. This was done by running the current plan model for 100 days and collecting the results; then adjusting the inter arrival times and lot sizes for each part in Witness, and collecting the results in the same way:

100 days	Small order average time	Small orders > 24 hours (%)	Large order average time	Large orders > 24 hours (%)	Total order average time	Total orders > 24 hours (%)
Current	6.047	0.00	52.329	71.12	9.604	5.47
Alternative	6.045	0.00	54.32	71.20	9.737	5.39

These results show what seems to be statistically insignificant differences in these measures between the current and alternative schedule. Small orders especially are not affected by the new arrival time as just like in the original model the stock levels never run out so there is little to no effect on the model. Again, showing how little this affects small orders is that both models maintain a 0.00% rate of small orders exceeding 24 hours, which shows excellent efficiency in handling small orders in both cases.

On the other hand, large orders show a different trend. The average processing time for large orders increased by approximately 2 hours in the alternative scenario, from 52.33 to 54.32 hours, coupled with a minor increase in the percentage of large orders exceeding 24 hours from 71.12% to 71.20%. This shows a slight decrease in efficiency for processing large orders in the alternative scenario.

Overall, while the total order average time saw a slight increase from 9.604 to 9.737 hours, the percentage of total orders exceeding 24 hours decreased slightly from 5.47% to 5.39%. Much like for small orders, changes this small show that the new delivery schedule barely affects the efficiency of the model.

These findings suggest that the newly developed model might slightly hinder certain aspects of the order processing, particularly in relation to large order completion times. The vast improvement in the delivery schedule not only as it keeps the same smoothness in operations as the previous schedule but moving from 15 separate deliveries a week being 7 for frames and 4 for both sheets and cushions to what could become less than 3 on average per week will ensure the company takes advantage of the static delivery costs of the parts instead of the other way around.

The improved delivery schedule also helps in improving how smooth the factory operated. This is shown by how the parts average 22 hours less from arrival to completion of assembly than in the standard model moving from 139hrs to 117hrs. Proving the factory operates more efficiently with this delivery schedule.

Average time to assembly. Old schedule

Name	No. Obs	Mean
PartTime	6015	139.047

Name	No. Obs	Mean
PartTime	6003	117.162

Average time to assembly. New schedule

Both plans take roughly the same amount of time for orders to be processed and shipped to the client, and the percentage of orders shipped within 24 hours also stays roughly the same. This means the new delivery schedule has a marginal impact on the efficiency of the manufacturing process itself, while providing a significant decrease of over 80% in delivery costs. Depending on the actual cost of each delivery, this could be a large reduction in operational costs per month for Congress and may be worth implementing for this reason.

All models have been prioritized for small orders over large ones. This is because of how the model is structured, and how the chairs are distributed into the warehouses.

Edit OUTPUT RULE FOR MACHINE Assembly

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```
IF Warehouse1Count > Warehouse2Count
 PUSH to Warehouse2
ELSE
 PUSH to Warehouse1
ENDIF
```

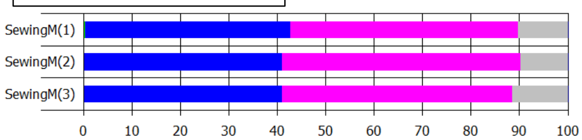
The output rule for the assembly machines pushes the chairs to the large order warehouse (Warehouse2), if the small order warehouse (Warehouse1) is fuller. Because of this, chairs are more likely to be sent to the small order warehouse, therefore small orders will be processed more quickly. This explains the 100% rate of small orders being completed within 24 hours, compared to the low number of large orders completed in 24 hours.

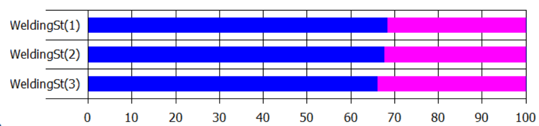
4)

When analyzing the model, you find that the key bottleneck revolves around the assembly element of the model. Due to the range of times in the distribution Gamma (2.74,11.2) the assembly times can range anywhere from around 5 minutes to 100 minutes with the average being around 30 minutes. The average alone takes significantly longer than any other machine in the model so when it hits the higher end of the distribution the assembly machines can end up becoming a large bottleneck for the model. This also shown by the buffers “SheetBuff2”, “FrameBuff” which are the buffers waiting on the assembly machines being full a lot of the time meaning that the machines before them “SewingM” for sheets and “WeldingSt” for frame being blocked often. The Pink bars showing when they are blocked.

Blue = Busy

Pink = Blocked





After looking at other potential improvements such as increasing the “FrameBuff” or switching the Warehouse storage method none of these quite had the impact of adding an extra assembly machine.

Adding one extra assembly machine speeds up the rate in which orders are completed dramatically. Dropping the average time an order spends in the model from 9.60 hours to just 7.28 hours.

Name	No. Obs	Mean	Standard Deviation	Min Value	Min Recorded...	Max Value	Ma Recor
TotalTimes	2439	7.282	7.994	0.037	77114.522	86.339	8140

And reducing the amount of orders that take over 24hrs to complete from 133 to 66. This leads to the percentage of orders that take over 24hrs dropping from 5.47% to 2.71%. Highlighting how much of an influence one extra assembly machine can affect the model's efficiency.

Name	No. Obs
TotalTime...	66

Due to how the assembled chairs are distributed into warehouses favoring the small orders, this extra assembly machine has little effect on the average time of small orders.

However, the large orders which are neglected in comparison with them averaging 52.3hrs in the model before completion, are impacted greatly by this new change with them dropping to below 24hrs in the model at 22.12hrs. A 30hr change with just one assembly machine, with the orders taking over 24hrs going from 133 to 66.

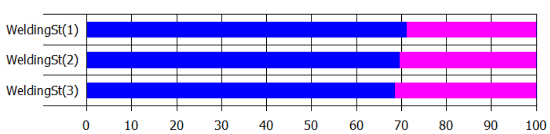
Name	No. Obs	Mean	Standard Deviation	Min Value	Min Recorded...	Max
LorderTim...	193	22.118	16.266	0.320	21221.558	

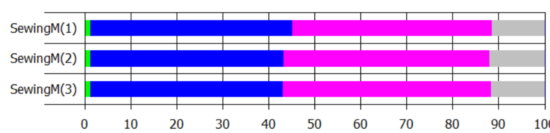
The model runs more efficiently than before up to the point of completion of the assembly process. With the average part whether it be Sheet, Frame or Cushion spending around 24hrs less in the model up to the being pushed into a warehouse than without the additional assembly machine.

Name	No. Obs	Mean
PartTime	6015	139.047

Name	No. Obs	Mean
PartTime	6226	114.915

In conclusion, the bottleneck which the large assembly times can be improved with adding an additional assembly machine. The blockage of the machines prior to assembly is improved, by around 5% less than previously, proving an improvement in the bottleneck. While at the same time vastly improving average times for Large orders which are one of the key problems in the current model even reducing them to on average less than 24 hrs.





Blue = Busy

Pink = Blocked

5)

The objective is to implement a quality control (QC) procedure after the assembly process at the Congress plant, and to evaluate its' impact. The QC step is expected to reduce the number of chairs being returned due to quality issues (which is currently about 6%), by checking the quality is up to standard before being shipped to the customer. Currently, the system does not have a QC procedure- chairs are being manufactured, stored in the warehouse, and shipped directly from the warehouse.

Congress faces a problem with the number of returned chairs, approximately 6%. The model needs to include a quality control step with a 3–10-minute inspection time, with a most likely value of 5 minutes. This could significantly impact production efficiency.

Adapting the model developed in Question 1, changes can be added to the model to ensure it accounts for the quality control step to increase production efficiency.

To implement this quality control step, the model now includes three new machines, called quality checkers, operated by new workers called quality checkers. These machines check the chair quality. The distribution used for the quality checker machines was the triangle distribution. This is because information about the quality control step is very limited and requires a simple distribution. There is minimal historical data about the distribution, but possess the knowledge about the maximum time, minimum time, and modal time. The triangular distribution allows knowledge to be incorporated into the model. The distribution used for the new machines is as follows;

Duration
Cycle Time:
Triangle (3,5,10)

Since the new elements distribution has been calculated, the code for the quality checker machines must be written. The code for each of the three quality checker machines shows as follows;

```
IF Warehouse1Count > Warehouse2Count
 PERCENT Warehouse2 94.00 ,SCRAP 6.00
ELSE
 PERCENT Warehouse1 94.00 ,SCRAP 6.00
ENDIF
```

This simulation code above manages chair inventory allocation between our two warehouses. It uses a decision rule dependent on current stock levels. If Warehouse1 has more chairs, the output focuses on Warehouse2. Additionally, 94% of chairs are kept and 6% are scrapped. However, if Warehouse2 has more chairs, the output focuses on Warehouse1 with the same percentages for keeping and scrapping. This code ensures that the warehouse with fewer chairs receives new stock, balancing inventory across both locations while considering potential defects in the chairs.

To develop a proper understanding of how the new model fluctuates with the new element allocations, it will be simulated for 100 days, or 133,510 minutes. This is to give an accurate representation of how the new changes to the model have influenced it. After running the model with the changes mentioned for this time, results are produced, which are visible in the table below. These results are then compared side by side to the results of the model produced for Q1, to show the difference in results after introducing the quality checkers, shown through spreadsheet results and histograms. To help explain the results and show where values come from, here is the TotalTimes and TotalTimes24 section from the standard model, which refers to the number of observations over the past 100 days:

Name	No. Obs	Mean	Standard Deviation	Min Value	Min Recorded At	Max Value	Max Recorded At
TotalTimes	2433	9.604	16.697	0.037	77114.522	146.129	133450.638

Name	No. Obs	Mean	Standard Deviation	Min Value	Min Recorded At	Max Value	Max Recorded At
TotalTimes24	133	68.195	31.487	24.090	106080.439	146.129	133450.638

Shown above is the Witness Total Observations for the model developed in the first question, and all the other information that is provided by the

Here are the tables with the same data, taken from the modified model with the QC step:

Name	No. Obs	Mean	Standard Deviation	Min Value	Min Recorded At	Max Value	Max Recorded At
TotalTimes	2430	10.670	21.454	0.040	126183.231	186.782	39437.940

Name	No. Obs	Mean	Standard Deviation	Min Value	Min Recorded At	Max Value	Max Recorded At
TotalTimes24	131	89.027	39.641	24.294	9242.066	186.782	39437.940

As clear in the data shown, both models have almost identical numbers of observations for 'TotalTimes', which indicates a consistent range of data collected across both the scenarios. This consistency is important as it enables us to make valid comparisons between the original model and the new one. This figure is random, so by ensuring that it is almost identical implies that the demand is still valid in our new model, and there are no mistakes

The increase in Mean for ‘TotalTimes’, from 9.60 to 10.67, suggests a slowdown in the process. This can be assumed to be because of the new Quality control step, which from the information present above, is proved to enhance the quality of the output. However, it seems to also slow down overall operations. The mean for orders that takes over 24 hours also increases significantly. It increases from 68.195 to 89.027, approximately a 31% increase. This could be due to the thoroughness required in quality checks, which may delay solving to orders that already have issues, leading to longer scrutiny and correction times.

The increase in standard deviation for both of our measured metrics in Question 5 implies introducing quality control brings more variability into the process. increased variability could be due to different times taken for quality checks depending on the issues detected, the variability in the correction processes, or the inconsistency in manual inspections. Now that we have this information and have shown an analysis of the tables, we can format it all into an excel table to put these values side by side and show exactly how they differ from each other.

100Days(133510 Time)	TotalAvTimes	TotalCount	Total>24	Percent>24	SorderAvTimes	SorderCount	Sorder>24	Percent>24	LorderAvTimes	LorderCount	Lorder>24	Percent>24
Standard	9.604	2433	133	5.47%	6.047	2246	0	0.00%	52.329	187	133	71.12%
Q5	10.669	2430	131	5.39%	6.057	2246	0	0.00%	66.965	184	131	71.20%

The figure above shows the comparisons between the standard model present in Question 1, and the model adapted for this question. AvTimes is the mean time for an order to be processed and shipped; Count is the number of chair orders completed in total, and for small orders (Sorder) and large orders (Lorder); >24 means the number and percentage of orders completed within 24 hours. These simulation results show that an addition of an extra element on the manufacturing chain paired with scrapping six percent of all chairs that come through does lead to some large changes to the efficiency and effectiveness of the model. To start with the average time of the total orders has an increase of an hour from 9.6hrs to 10.6hrs this is to be expected due to the extra time that it takes for the chairs to be pushed into the warehouses and 6% of chairs never making it to the warehouse. Although the percentage of orders over 24hrs does take a very minuscule drop this is only due to the model not completing some of the orders which it did in the standard model.

As mentioned earlier the small orders are favorited in the model. This means that the small order times are not affected to any significant degree by the implementation of the quality checking machines. Although the large orders feel the full effect of the new addition with orders more than half a day longer on average at 14.6 extra hours added onto the orders.

In the 100 days of operation, the model assembled 6162 chairs in total, and of these 369 chairs were scrapped, this can be found by looking at the number of operations of the Quality Check machines and comparing with the total in from both warehouses. Obviously 369 chairs not being shipped is going to affect the model's efficiency and this is shown by greater average order completion times, but the quality check element is obviously not an element meant to maximize efficiency and instead meant to tackle the problem with sending out faulty products. Therefore, in a real world scenario having quality check will be useful as the standard model does not equate for orders being returned which can lead to more problems for the factory and a quality check will be able to fix this problem.